NEW RADIO COMMUNICATION FOR NEW ERA

NRC7394 Evaluation Kit User Guide

(Transmit Power Control (TPC))

Ultra-low power & Long-range Wi-Fi

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NRC7394 Evaluation Kit User Guide (Transmit Power Control (TPC))
Ultra-low power & Long-range Wi-Fi

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1 Overview and prerequisites

TPC (Transmit Power Control) adjusts the transmit power of a STA automatically according to the link condition. It lowers the power when the link has margin and raises it again when more power is needed. Enable the feature, then verify its operation.

TPC requires firmware that supports it and an upstream AP that supports it. It is disabled by default. TPC is a Newracom proprietary feature. It works only between NRC7394 devices that both have TPC enabled; with a third-party AP, the STA keeps its normal transmit power.

Table 1.1 TPC behavior by device role

Device role	Behavior when TPC is enabled
AP	Advertises TPC support to the STAs. The AP's own transmit power is not lowered
STA	Checks the upstream AP's support, then adjusts its own transmit power automatically
RELAY	Adjusts only the STA-side power towards the upstream AP. The AP-side power used to serve downstream STAs is unchanged
MESH	Not supported

How much the power drops depends on the link condition: distance, interference and the transmission rate. Enabling the feature does not always lower the power. The power reduction that TPC reports is not a battery-life or power-consumption saving figure.

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2 Quick start

2.1 Using the Web GUI

The Web GUI is a reference tool; see UG-7394-032-Web_GUI.

1. Open the Web GUI of the upstream AP.
2. Under Settings → Radio, set Transmit power control to on, press Apply, and follow the restart notice.
3. Enable TPC on the STA in the same way and apply the setting.
4. After the wireless service restarts, confirm that the STA reconnects so that both sides exchange their TPC support information.
5. Send data from the STA towards the upstream AP and check the TPC state on the dashboard.

Changing TPC in the Web GUI requires a full wireless restart when the wireless function is running. Follow the application notice and wait for reconnection. A saved setting is applied again the next time the wireless function starts.

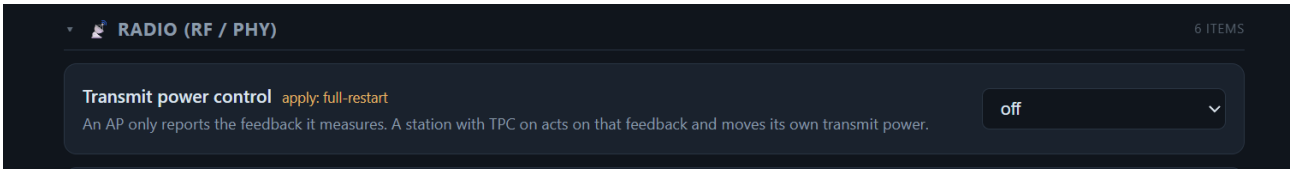


Figure 2.1 TPC selection under Radio

2.2 Using the CLI

Start the wireless function on the AP and on the STA, then run the following commands on each.

```
cd ~/nrc_pkg/script
./cli_app set tpc on
./cli_app set tpc show
```

Reassociate a STA that is already connected after TPC is enabled. Confirm that both `TPC` and `Peer TPC` are `on` on the STA, then send uplink traffic.

A setting made from the CLI does not survive a firmware restart. To keep the setting, use the Web GUI, or run the command again every time the wireless function starts.

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3 Configuration

3.1 Turning TPC on and off

Table 3.1 TPC CLI commands

Task	CLI command
Enable	`cli_app set tpc on`
Disable	`cli_app set tpc off`
Check the enabled state	`cli_app set tpc show`
Check the power adjustment result	`cli_app set tpc stats`

`on` and `off` are enough for normal use. The firmware performs the power adjustment; there is no need to set a power value manually.
When starting the wireless function with start.py, set tpc = 1 in its header on both the AP and STA before startup (0 disables TPC).

```
#------------------------------------------------------------------------------#
# TX Power Control (AP:Feedback only   STA:Control TX POWR according to Feedback from AP)
tpc = 1                       # 0 (disable) or 1 (enable)
#------------------------------------------------------------------------------#
```

Figure 3.1 TPC startup setting in start.py

3.2 Scope and cautions

- `TPC: on` on an AP means the AP advertises support to the STAs. Do not read it as the AP lowering its own power.
- A STA adjusts its power only after the upstream AP's TPC support has been confirmed. The usual way to confirm it is to enable the feature on both sides and reassociate the STA.
- The power may stay high when the link condition is poor or a low transmission rate is in use. In weak-signal conditions TPC converges on maximum power by design.
- The power is adjusted within the range the board data allows and within the configured TPC power range.

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4 Checking operation

4.1 Checking the enabled state

```
./cli_app set tpc show
```

Check these three items in the output.

Table 4.1 Items in the set tpc show output

Item	How to read it
`Role`	Whether the device is currently AP, STA or RELAY
`TPC`	`on` when the local feature is enabled
`Peer TPC`	The upstream AP's TPC support as seen by a STA. `off` is normal on an AP without a STA interface

For example, an AP shows the following even with TPC enabled.

```
Role                          : AP
TPC                           : on
Peer TPC                      : off
OK
```

This output means the AP is advertising TPC support. It does not report any power adjustment on the AP itself.

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4.2 Checking the actual power adjustment

On firmware that supports the statistics, run the following on the STA.

```
./cli_app set tpc stats clear
```

Send data from the STA towards the upstream AP, then read the statistics.

```
./cli_app set tpc stats
```

The following items are enough to start with. If collection is off, run `./cli_app set tpc stats on` first.

Table 4.2 Main items in the set tpc stats output

Item	Meaning
`txdone`	Number of transmit-completion events TPC processed. Check that it increases after traffic is sent
`txp`	Applied power: latest value (`last`), average (`avg`), minimum and maximum, in dBm
`saving avg`	Average reduction against the board-data power, in dB
`ack_rate`	Ratio of events processed by TPC that received an ACK
`retry`	Distribution by retry count. If retries increase, check the link condition as well

`txp` is the power the firmware tracks, not an RF measurement. The statistics count only the events TPC processed, so they are not the same as the delivery ratio of the whole network. Confirm that actual data traffic is healthy alongside the power reduction.

```
pi@raspberrypi:~/nrc_pkg/script $ sudo ./cli_app set tpc show
Role                            : STA
TPC                             : on
Peer TPC                        : on
OK
pi@raspberrypi:~/nrc_pkg/script $ sudo ./cli_app set tpc stats
tpc: 0 STATS ---
 state_enter NONE:0 INIT:2 WAIT:1 READY:4217 SET:4216
 init_reason initial:1 no_ack:0 hi_retry:0 wait_noack:1
 txdone 13555  no_ack 0  ack_rate 100.0%
 retry  0:13521 1:17 2:17 3:0 4+:0
 snr    samples:1 last:31 avg:31 min:31 max:31
 txp    samples:13554 last:10 avg:10 min:10 max:19
 txp_hist 10:13176 11:90 12:90 13:91 14:20 15:20 16:21 17:19 18:10 19:17
 floor_hit:13176 (97.2%)  ceil_hit:0 (0.0%)
 saving avg:8.9 dB  total:120882 dBm-pkt
 apply  total:4217  cand:1 cand_up:0 retry_up:17 stable_down:416 stable_keep:3783, low_mcs:0 rst_noack:0 rst_hretry:0
OK
```

Figure 4.1 Example TPC state and power adjustment statistics on a STA

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5 Troubleshooting

Table 5.1 TPC troubleshooting

Symptom	What to check and do
TPC enabled on the AP but the power does not drop	This is normal. The actual power adjustment happens on the STA side.
`Peer TPC` is `off` on the STA	Check that the upstream AP supports TPC and has it enabled, then reassociate the STA.
Enabled on the STA but the power does not drop	Send uplink traffic and read the statistics. A poor link or a low transmission rate can keep the power high.
All statistics are zero	Check that you are reading them on the STA, that uplink traffic exists, and that both TPC and statistics collection are on.
`tpc: 0 stats is off` is shown	Turn collection on with `./cli_app set tpc stats on`.
The `stats` command is not supported	Check whether the firmware includes the statistics feature. Its presence is independent of whether TPC is enabled.
`MESH does NOT support TPC` is shown	The feature cannot be used in mesh mode.
TPC is off after a restart	A CLI setting is not retained. Set it in the Web GUI, or apply it again after startup.
Retries increase or the link becomes unstable	Check the signal and interference. If needed, disable TPC, compare under the same conditions and send the result to technical support.

When you contact technical support, include the device role, the firmware version, the output of `set tpc show` and `set tpc stats`, and the traffic conditions under which the problem occurred.

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6 Advanced commands

Normal use does not require any of the settings in this section.

6.1 Enabling without reassociation

```
./cli_app set tpc on -f
```

Use this only when you have already confirmed that the upstream AP supports TPC and that the feature is enabled on both sides. The command sets the local support flag and skips the exchange of support information. It does not add TPC support to the upstream AP.

After the command, `Peer TPC: on` is a value set by the command. Do not read it as a result confirmed through an actual exchange of support information. This option cannot be used in mesh mode either.

6.2 Managing the statistics

Table 6.1 Statistics management commands

Command	Behavior
`./cli_app set tpc stats clear`	Clears the accumulated counters
`./cli_app set tpc stats on`	Starts or resumes collection
`./cli_app set tpc stats off`	Stops collection and clears the accumulated counters

Collection is on by default. `off` is not a pause that preserves the existing statistics.

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7 Revision history

Version	Date	Description
Ver 1.0	9/28/2026	Initial release

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